

| Phase   | Task              | Metric                  | Group means |        |       |         |             |       | Pairwise comparison ( $p$ value) |
|---------|-------------------|-------------------------|-------------|--------|-------|---------|-------------|-------|----------------------------------|
|         |                   |                         | Humans      | Senior | PhD   | Experts | Non-Experts | GenAI |                                  |
| Pre-ML  | Racial inequality | No. of paths            | 5.18        | 5.17   | 5.19  | 5.20    | 5.18        | 6.64  | 0.014*                           |
|         |                   | Maximum path length     | 2.14        | 2.00   | 2.17  | 1.87    | 2.21        | 2.64  | 0.004**                          |
|         |                   | No. of latent variables | 2.15        | 2.00   | 2.19  | 1.67    | 2.29        | 5.16  | < 0.001***                       |
|         | Gender inequality | No. of paths            | 5.24        | 5.23   | 5.24  | 5.12    | 5.27        | 6.80  | 0.027*                           |
|         |                   | Maximum path length     | 2.03        | 2.08   | 2.02  | 2.00    | 2.04        | 2.60  | 0.001**                          |
|         |                   | No. of latent variables | 1.94        | 1.85   | 1.97  | 1.62    | 2.04        | 5.52  | < 0.001***                       |
| Post-ML | Racial inequality | No. of paths            | 5.09↓       | 4.25↓  | 5.26↑ | 4.40↓   | 5.28↑       | 8.28↑ | 0.002**                          |
|         |                   | Maximum path length     | 2.00↓       | 1.75↓  | 2.05↓ | 1.80↓   | 2.06↓       | 2.68↑ | 0.001**                          |
|         |                   | No. of latent variables | 2.03↓       | 2.08↑  | 2.02↓ | 1.87↑   | 2.07↓       | 6.72↑ | < 0.001***                       |
|         | Gender inequality | No. of paths            | 5.11↓       | 4.77↓  | 5.19↓ | 4.69↓   | 5.23↓       | 8.56↑ | 0.010*                           |
|         |                   | Maximum path length     | 1.99↓       | 1.85↓  | 2.02  | 1.88↓   | 2.02↓       | 2.60  | 0.001**                          |
|         |                   | No. of latent variables | 1.94        | 1.54↓  | 2.03↑ | 1.38↓   | 2.11↑       | 7.28↑ | < 0.001***                       |